

# Alex Gower

Math & Algorithms Group, Nokia Bell Labs and Theoretical Condensed Matter Group, University of Cambridge

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## EDUCATION

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**Nokia Bell Labs and University of Cambridge, Jesus College**

Sep 2022 – Present

*PhD, Project Title: “The Physics of Machine Learning on Oscillator Ising Machines”*

**University of Oxford, Lincoln College**

Oct 2018 – Jun 2022

*MMathPhys: Master of Mathematical and Theoretical Physics, Distinction (73%)*

*BA: Physics, First Class (82%, Ranked 1st in Cohort of ~200 in Second Year Examinations)*

## TECHNICAL SKILLS

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**Programming:** Python, Julia, Java, MATLAB, FORTRAN, R, C++

**Libraries:** PyTorch, TensorFlow, XGBoost, LightGBM, InterpretML, NumPy, Pandas, Hyperopt, TransformerLens

**Technologies:** Django, Android Studio, Git, SLURM, Hugging Face, Jupyter, Google Colab, Bash, AWS

## RELEVANT EXPERIENCE

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**Nokia Bell Labs and University of Cambridge**

*PhD Researcher*

*Sep 2022 – Present*

- Researching Equilibrium Propagation and energy-based machine learning, spanning analog neuromorphic hardware (Oscillator Ising Machines), extensions to generative modelling (flow matching), and theoretical analysis of glassy dynamics on rugged energy landscapes

*Math & Algorithms Group*

*Nokia Bell Labs*

- Introduced Gradient Equilibrium Propagation (GradEP), extending EP to train energy gradients rather than energy minima, and demonstrated the first EP-trained flow matching generative model (FlowEqProp) — **Best Paper Award, ICONS '26**
- Developed a novel EP learning algorithm for Oscillator Ising Machines (MLP and CNN architectures), achieving SoTA neuromorphic accuracy on MNIST/FashionMNIST while maintaining hardware-constraint resilience (*ICONS 2025, NeurIPS 2025 ML4PS*); now being implemented on hardware with Nokia and Eindhoven University
- Researching extensions of EP to more expressive energy-based architectures (modern Hopfield networks, energy transformers) for scaling EP-trained generative models
- Engineered OIM performance for combinatorial optimization problems (Max-SAT, Max-Cut, spin-glass models) using dimensional annealing and memory dynamics algorithms
- Collaboratively developed a GNN model to learn optimal Max-SAT clause weights for combinatorial optimisation
- Optimized large-scale ML simulations by implementing multi-threaded Julia/PyTorch hybrid processing pipeline with custom memory management, reducing computational runtime by 10x on distributed HPC clusters

*Theoretical Condensed Matter Group*

*University of Cambridge*

- Authored paper on rugged energy landscapes (submitted to *Nat. Phys.*) in which slow gradient descent dynamics on a toy energy (loss) landscape are analysed, uncovering a saddles-to-minima topological crossover in the critical point connectivity which is suppressed (and fast dynamics recovered) upon the addition of ‘swap moves’
- Supervised a Master’s student dissertation project investigating saddles-to-minima topological crossovers and high dimensional route suppression in kinetically constrained Ising models
- Supervised Statistical Mechanics and Mathematical Methods undergraduate course classes for 3 years

**Alan Turing Institute & UK Ministry of Justice**

*Machine Learning Technical Project Lead*

*Aug 2025 – Present*

- Built short-horizon violence risk models (LightGBM, XGBoost, EBM) using 1,500+ engineered features across 20M+ records, improving recall by 50% over prior baselines and achieving state-of-the-art short-term violence forecasting performance
- Built reduced-feature interpretable EBM models suitable for high-stakes decision-making, and surfaced actionable risk factors from feature importances to inform frontline operations
- Built several MoJ products using LLM pipelines for semantic incident categorisation and case note surfacing

## MARS Alignment Research Programme

Research Fellow

Cambridge AI Safety Hub

Dec 2024 – May 2025

- Completed research programme evaluating frontier LLM abilities in model compression, mechanistic interpretability and feature steering for ML R&D tasks using InspectAI framework

## ADDITIONAL TRAINING

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### ELLIS Cambridge Summer School on Probabilistic Machine Learning

Selected Participant

University of Cambridge

Jul 2026

- Selected for competitive week-long summer school covering probabilistic modelling, uncertainty quantification in deep learning, generative models, and decision-making under uncertainty; presented at poster session

### Machine Learning Workshops

Workshop Attendee

G-Research

Apr 2025

- Completed G-Research ML workshops on gradient boosting, XGBoost, and quantitative finance applications

### MPhil Machine Learning and Machine Intelligence

PhD Student Auditor

University of Cambridge

Sep 2023 – Present

- Completed courses in the Machine Learning and Machine Intelligence Master's course including Probabilistic and Bayesian Machine Learning, Computational Statistics and Deep Learning and Structured Data

### Cambridge AI Safety Hub

Fellowship Graduate

University of Cambridge

Sep 2024 – Sep 2025

- Completed intensive AI safety fellowship with projects in LLM interpretability, covering topics including transformers, RLHF, RAG, mechanistic interpretability, and agent architectures

## PROJECTS

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### itsgiving | [Website](#)

2025

- Built NLP semantic college matching system using custom word embeddings and by fine-tuning a HuggingFace LLM DistilBERT, enabling natural language queries to match prospective students with colleges based on social media language data

### Odin | [Android App Store](#)

2016

- Developed 3 language learning and flashcard apps (20K+ downloads) using Java and Android Studio; won national 'Apps for Good' competition

### Oppia | [Website](#)

2016

- Contributed to 'Oppia' (an open-source education website) while mentored by Google software engineer

## ADDITIONAL RESEARCH EXPERIENCE

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### Quantum Matter in High Magnetic Fields Group

Undergraduate and Graduate Research Assistant:

University of Oxford

Jun 2021 – Aug 2022

- Co-authored two Physical Review B papers on multi-band superconductivity and impurity effects in FeSe, developing theoretical models and a MATLAB pipeline (tight-binding, multi-band, DFT) to extract band parameters from quantum oscillation and upper critical field experiments

### ISIS Neutronics Group

Nuclear Analysis Summer Student

Science Technology and Facilities Council

Jun 2020 – Sep 2020

- Enhanced FETCH2 radiation modeling codebase (following unit and continuous integration testing practices); corrected critical cross-section handling errors and presented benchmarking results at Imperial College conference

## PUBLICATIONS

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- **Alex Gower**. FlowEqProp: Training Flow Matching Generative Models with Gradient Equilibrium Propagation. In *Proc. of ICONS '26*, p60–66. IEEE Press, 2026. doi:10.1145/3822454.3822485 [**Best Paper Award**]
- **Alex Gower**. Learning at the Speed of Physics: Equilibrium Propagation on Oscillator Ising Machines. Presented at NeurIPS 2025: Machine Learning and the Physical Sciences (ML4PS) Workshop. arXiv:2510.12934
- **Alex Gower**. How to train an Oscillator Ising Machine using Equilibrium Propagation. In *Proc. of ICONS '25*, p229–234. IEEE Press, 2025. doi:10.1109/ICONS69015.2025.00042
- **Alex Gower**, Oliver Hart, and Claudio Castelnovo. Saddles-to-minima topological crossover and glassiness in the Rubik’s Cube, Oct 2024. arXiv:2410.14552
- Z. Zajicek, S. J. Singh, H. Jones, P. Reiss, M. Bristow, **A. Gower** et al. Drastic effect of impurity scattering on the electronic and superconducting properties of Cu-doped FeSe. *Physical Review B*, 105(11):115130, Mar 2022.
- M. Bristow, **A. Gower**, A. I. Coldea et al. Multiband description of the upper critical field of bulk FeSe. *Physical Review B*, 108(18):184507, Nov 2023.

## OTHER SKILLS AND INTERESTS

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Acrobatic Gymnastics (Cambridge University Double Blue), Jesus College Graduate Social Officer, Charity (LincZambia)